Quantification: I534_z19c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	37
interior lacuna count	29
frame-edge lacuna count	8

Interior lacunae only: mean, median and sample SD (n = 29)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2639.93	1712	2160.07	px^2
major_axis_length_px	lacuna shape	81.1487	84.2608	30.5482	px
minor_axis_length_px	lacuna shape	43.5485	38.1837	18.3642	px
aspect_ratio	lacuna shape	1.9739	1.9701	0.5912	unitless
eccentricity	lacuna shape	0.7986	0.8616	0.1548	unitless
solidity	lacuna shape	0.7476	0.7469	0.0929	unitless
roots_count	headline	9.7931	9	4.7236	count
ring_length_r30_px	headline	257.103	249	99.9094	px
ring_length_r60_px	reported	473.552	474	192.629	px
owned_length_px	ownership-dependent	515.732	404.605	327.795	px
edge_count	ownership-dependent	25.5172	21	14.9059	count
mean_edge_length_px	ownership-dependent	19.4355	19.2669	3.9006	px
perimeter_px	normalised	260.748	272.936	114.29	px
ring_area_r30_px2	normalised	8479.14	8089	3367.3	px^2
ring_area_r60_px2	normalised	16657.2	17440	6531.99	px^2
in_frame_fraction_r30	normalised	0.9979	1	0.009	unitless
in_frame_fraction_r60	normalised	0.9851	1	0.0446	unitless
ring_density_r30	normalised	0.0320683	0.0308512	0.0119741	px^-1
ring_density_r60	normalised	0.0292887	0.0276852	0.00964869	px^-1
roots_per_100px_perimeter	normalised	4.1044	3.896	1.8346	per 100 px
ring_attached_length_r30_px	network v2	228.793	230	95.6248	px
ring_attached_length_r60_px	network v2	356	378	151.636	px
ring_length_w_r30_px	network v2	292.221	280.037	114.871	px
ring_length_w_r60_px	network v2	538.565	525.433	220.872	px
sholl_crossings_r10	network v2	5.6897	5	2.674	count
sholl_crossings_r20	network v2	6.5172	7	3.4705	count
sholl_crossings_r30	network v2	6.2759	7	3.2502	count
loop_count	loops, report only	0.931	0	1.387	count
loop_edge_share	loops, report only	0.0856	0	0.1204	share
ring_width_mean_r30_px	width	7.5364	7.4938	0.9121	px

Per field
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measure	value	units
analysed_area_px2	885177	px^2
total_skeleton_length_px	23531	px
canalicular_length_density_per_px	0.0265834	px^-1
junction_count	477	count
junction_density_per_px2	0.00053888	px^-2
lacuna_count	37	count

lacunae_per_px2	4.18e-05	px^-2
skeleton_component_count	309	count
mean_component_length_px	76.1521	px
median_component_length_px	27	px
field_length_density_w_per_px	0.030232	px^-1
field_density_without_flagged_per_px	0.0277474	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	471	px
unattached_component_count	28	count
canalicular_length_density_attached_per_px	0.0265834	px^-1
field_length_density_with_unattached_per_px	0.0271155	px^-1
loop_count	38	count
loop_edge_share	0.1067	share
edge_count	1265	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.0926	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	22815	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	7213	156.74	77.5309	2.0216	0.8691	0.6864	-1.375	29.9816	225.264	True
2	1583	53.2276	47.8198	1.1131	0.4392	0.7794	0.9204	26.8503	401.312	True
3	602	44.8433	24.2407	1.8499	0.8413	0.7516	-1.0436	9.8123	650.007	True
4	12484	376.172	54.3639	6.9195	0.9895	0.6519	1.5686	31.5111	841.265	True
5	2878	110.538	36.55	3.0243	0.9438	0.8432	-1.5019	47.2293	619.351	False
6	1304	60.2003	30.9259	1.9466	0.858	0.7893	-1.2145	90.1833	146.716	False
7	1823	56.4889	49.8735	1.1326	0.4696	0.8271	-1.1579	125.003	757.529	False
8	19030	208.347	131.926	1.5793	0.774	0.7439	-1.1645	208.508	86.0397	True
9	3354	87.3099	52.7612	1.6548	0.7968	0.8689	-1.2343	222.521	989.166	True
10	10008	137.275	97.443	1.4088	0.7044	0.8833	-1.0949	269.479	692.84	False
11	1637	57.8693	48.4179	1.1952	0.5477	0.7276	-0.0784	295.51	896.101	False
12	1291	47.4215	38.1837	1.2419	0.593	0.8741	0.4227	324.111	336.189	False
13	627	51.449	19.4729	2.6421	0.9256	0.7093	0.4876	337.973	69.1467	False
14	6026	129.945	61.3358	2.1186	0.8816	0.8485	-1.1185	377.349	511.866	False
15	470	46.383	15.4885	2.9947	0.9426	0.7309	-1.0302	386.604	245.745	False
16	1342	66.6735	36.4005	1.8317	0.8378	0.6286	1.399	396.13	787.283	False
17	1283	64.6772	37.6197	1.7192	0.8134	0.6386	1.1702	434.174	798.524	False
18	2978	75.2858	55.7926	1.3494	0.6714	0.7831	-0.8753	453.94	606.7	False
19	596	38.0384	25.7402	1.4778	0.7363	0.7469	-0.9961	464.5	212.371	False
20	4876	84.2608	78.5547	1.0726	0.3617	0.7525	0.8631	503.31	62.7231	False
21	3737	84.848	59.7745	1.4195	0.7097	0.8338	-0.4377	493.364	441.851	False
22	820	51.4532	24.5895	2.0925	0.8784	0.7614	1.0688	505.329	211.1	False
23	3553	104.824	44.321	2.3651	0.9062	0.8721	0.2384	581.822	856.943	False
24	4782	122.937	62.4014	1.9701	0.8616	0.6511	1.4416	592.132	630.596	False

25	1615	90.9802	31.7451	2.866	0.9372	0.5886	-0.9716	606.27	724.946	False
26	2632	101.535	38.4077	2.6436	0.9257	0.6745	-1.3449	587.391	245.703	False
27	5166	140.866	61.0892	2.3059	0.9011	0.6428	0.013	686.162	69.5337	False
28	1712	87.747	36.8261	2.3827	0.9077	0.6786	-0.5403	724.265	671.19	False
29	914	54.9878	33.7201	1.6307	0.7899	0.5745	-0.9811	788.83	430.015	False
30	2692	86.6013	40.7838	2.1234	0.8822	0.8628	-1.1084	820.32	99.4658	False
31	1680	84.8432	32.3344	2.6239	0.9245	0.7391	-0.8417	827.977	860.805	False
32	4209	112.297	51.8973	2.1638	0.8868	0.8336	-0.6033	855.455	275.629	False
33	46700	513.831	176.501	2.9112	0.9392	0.632	-1.5581	964.214	759.102	True
34	681	43.231	24.4304	1.7696	0.825	0.8107	-1.479	862.358	1007.22	True
35	627	40.588	26.12	1.5539	0.7654	0.6883	1.4019	907.663	832.011	False
36								92.8281	393.154	False
37								194.154	525.221	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	16	358	539	385.433	27	14.2753	7.4611
2	3	181	349	240.924	14	17.2088	7.1861
3	4	31	31	32.9706	2	16.4853	3.1253
4	7	250	471	167.51	13	12.8854	7.9985
5	3	116	288	38.2132	2	19.1066	8.5717
6	6	126	126	158.225	10	15.8225	5.6165
7	5	175	382	404.605	21	19.2669	7.2624
8	24	700	995	1166.79	62	18.8192	7.2906
9	3	105	193	168.711	8	21.0888	6.6191
10	18	485	783	851.323	45	18.9183	8.0435
11	7	132	249	259.936	14	18.5669	6.7288
12	9	263	637	1186.99	51	23.2743	6.8092
13	6	137	295	88.0833	5	17.6167	5.1225
14	20	382	644	554.931	36	15.4148	7.0898
15	6	267	436	873.708	31	28.1841	7.0799
16	8	239	474	611.203	31	19.7162	7.5543
17	13	333	624	900.666	48	18.7639	7.1833
18	9	239	330	357.635	20	17.8817	8.132
19	10	242	307	353.179	29	12.1786	7.4692
20	16	421	621	830.909	34	24.4385	7.4938
21	15	237	506	307.764	17	18.1038	7.4688
22	10	267	303	244.439	14	17.4599	7.9574
23	15	328	776	1169.86	54	21.6641	8.1675
24	14	312	526	374.635	19	19.7176	7.9321
25	7	176	268	120.125	7	17.1607	7.936
26	17	368	607	475.86	21	22.66	7.4204
27	14	404	623	335.735	17	19.7491	7.9182
28	9	332	601	752.097	34	22.1205	7.1655
29	12	249	609	729.027	32	22.7821	7.5833
30	8	236	400	298.392	15	19.8928	6.9221
31	6	249	424	399.291	17	23.4877	8.1693
32	9	273	695	582.014	24	24.2506	6.8395
33	24	556	856	880.921	44	20.0209	8.3869
34	3	108	142	173.439	9	19.271	8.5687
35	3	47	47	23.8995	3	7.9665	10.0157
36	3	226	424	733.127	42	17.4554	8.3447
37	6	195	728	940.352	47	20.0075	8.5572

Normalised per lacuna

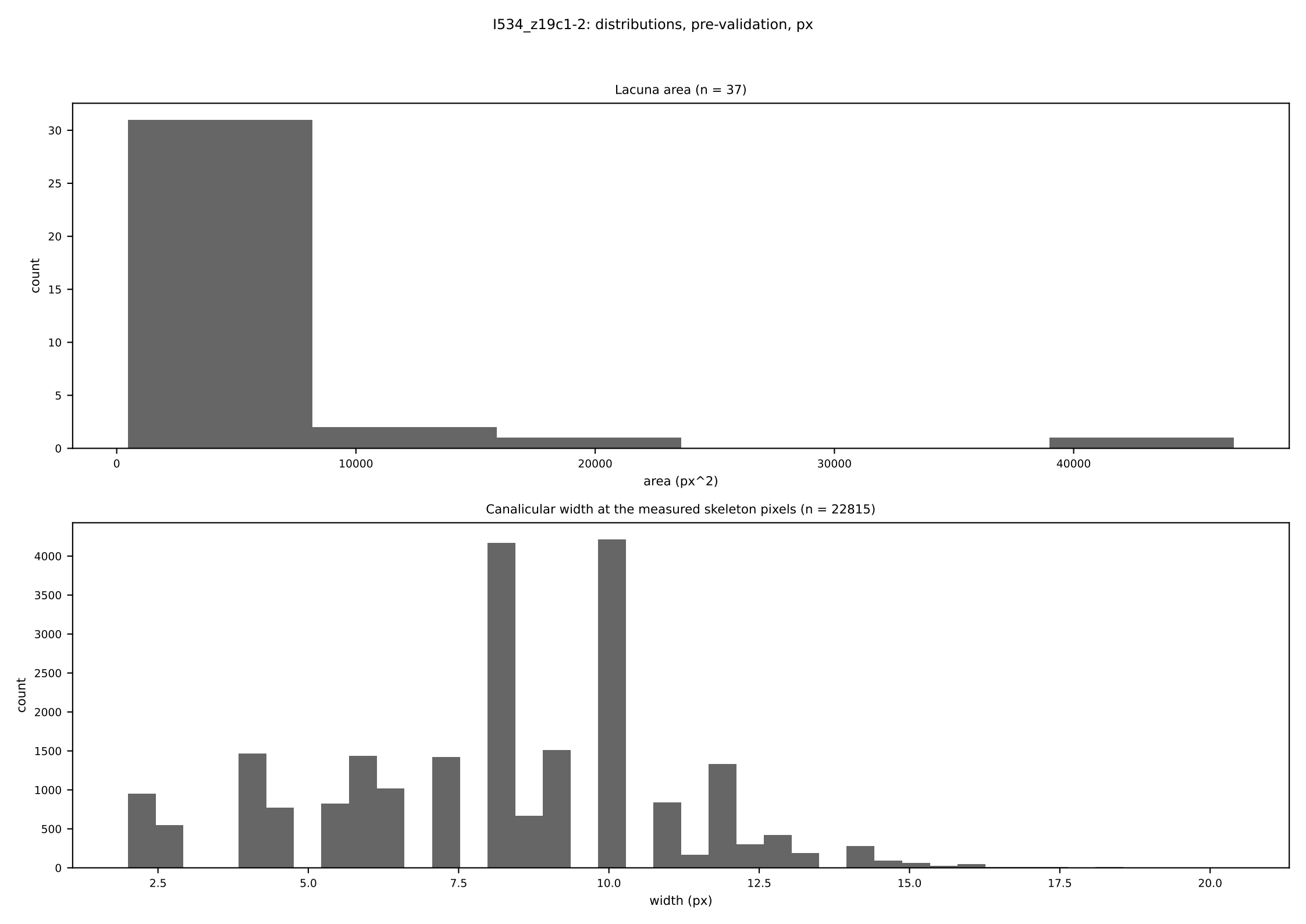
lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter

1	685.305	9981	15387	0.6811	0.6379	0.0358682	0.0350296	2.3347
2	195.823	5712	9539	0.7604	0.6834	0.0316877	0.0365866	1.532
3	151.64	1114	1114	0.604	0.5695	0.0278276	0.0278276	2.6378
4	1095.21	15947	22773	0.64	0.5463	0.0156769	0.0206824	0.6391
5	287.179	7182	14547	0.9855	0.8408	0.0161515	0.0197979	1.0446
6	193.823	4653	6512	1	1	0.0270793	0.0193489	3.0956
7	272.936	8089	13798	1	1	0.0216343	0.0276852	1.8319
8	1123.86	18396	29950	0.8124	0.7656	0.0380518	0.033222	2.1355
9	266.788	7353	14089	0.7107	0.6653	0.0142799	0.0136986	1.1245
10	462.008	15283	28784	1	1	0.0317346	0.0272026	3.896
11	283.463	8651	18027	1	1	0.0152584	0.0138126	2.4695
12	164.51	7220	17440	1	1	0.0364266	0.0365252	5.4708
13	144.539	5829	11980	1	0.9955	0.0235032	0.0246244	4.1511
14	382.854	12834	21714	1	1	0.0297647	0.0296583	5.2239
15	123.539	6120	12798	1	1	0.0436274	0.0340678	4.8568
16	262.871	6200	11714	1	1	0.0385484	0.0404644	3.0433
17	296.191	5917	12004	1	1	0.0562785	0.0519827	4.3891
18	274.735	10367	18090	1	1	0.0230539	0.0182421	3.2759
19	132.539	4228	7553	1	1	0.0572375	0.0406461	7.5449
20	375.103	12127	20489	0.9532	0.8444	0.0347159	0.030309	4.2655
21	300.049	10385	21896	1	1	0.0228214	0.0231092	4.9992
22	155.717	5059	7404	1	1	0.0527772	0.0409238	6.4219
23	288.735	10625	22862	1	1	0.0308706	0.0339428	5.1951
24	439.83	12113	21276	1	1	0.0257575	0.0247227	3.183
25	289.848	8213	14035	1	1	0.0214294	0.0190951	2.4151
26	316.777	10912	23620	1	1	0.0337243	0.0256986	5.3666
27	533.263	15650	25399	1	0.8882	0.0258147	0.0245285	2.6253
28	263.664	9930	20913	1	1	0.033434	0.0287381	3.4134
29	215.723	8071	20618	1	1	0.0308512	0.0295373	5.5627
30	247.279	9530	20421	1	0.9977	0.0247639	0.0195877	3.2352
31	251.22	7989	14551	1	1	0.0311679	0.0291389	2.3883
32	352.617	11633	27463	1	1	0.0234677	0.0253068	2.5523
33	2793.18	34686	49456	0.656	0.5847	0.0160295	0.0173083	0.8592
34	125.154	3865	6518	0.681	0.6229	0.0279431	0.0217858	2.397
35	130.468	2567	2567	1	1	0.0183093	0.0183093	2.2994
36	57.9411	3975	9943	1	1	0.0568553	0.0426431	5.1777
37	62.2843	4543	14641	1	1	0.0429232	0.0497234	9.6333

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	278	398	412.161	613.843	6	7	8
2	114	245	203.853	396.877	4	7	5
3	27	27	34.5563	34.5563	0	0	0
4	169	169	277.137	526.617	10	6	6
5	61	167	129.397	324.664	3	4	5
6	121	121	140.397	140.397	3	1	0
7	164	309	201.823	449.588	5	8	8
8	684	837	797.24	1126.86	15	13	9
9	59	101	122.811	220.409	1	1	4
10	419	622	550.943	893.607	14	14	11
11	108	149	146.196	272.38	4	1	4
12	213	436	295.865	726.926	4	9	9
13	122	132	159.338	341.433	4	1	2
14	326	445	428.433	725.299	8	12	10
15	267	398	308.764	503.688	4	4	7
16	221	384	265.238	525.433	5	8	6

17	306	536	385.433	714.269	7	7	8
18	230	283	276.764	375.777	5	6	3
19	206	245	280.037	343.764	3	7	4
20	409	556	486.931	705.227	10	11	10
21	222	282	265.794	568.973	6	5	4
22	258	264	296.995	333.894	5	8	2
23	304	568	365.007	873.068	7	9	13
24	286	345	346.664	597.6	6	3	5
25	176	256	199.995	306.735	4	3	4
26	317	405	418.647	691.512	8	10	7
27	343	430	460.546	711.583	9	11	10
28	306	510	389.505	691.955	8	9	9
29	249	486	288.836	691.127	5	4	5
30	199	250	270.007	457.517	6	8	7
31	245	378	275.823	475.676	8	6	5
32	238	390	320.706	811.193	7	7	8
33	443	640	622.813	965.205	17	16	12
34	75	75	123.64	154.125	2	3	1
35	40	40	48.9706	48.9706	1	1	0
36	199	358	256.166	487.818	3	7	7
37	80	579	215.196	828.311	3	5	9


I534_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 37)
count
area (px^2)
0
5
10
15
20
25
30
0
10000
20000
30000
40000
Canalicular width at the measured skeleton pixels (n = 22815)
count
width (px)
0
500
1000
1500
2000
2500
3000
3500
4000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0